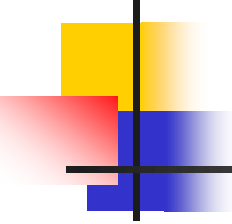


# Physics (PHYS-1101)

## THERMAL PROPERTY OF MATTER

### Lecture 9:

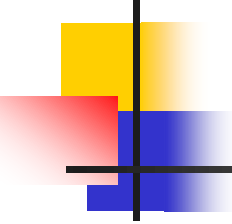
### Energy in Thermal Processes



# Energy Transfer, $Q$

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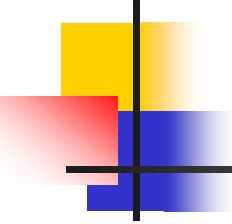
- When two objects of different temperatures are placed in thermal contact, the temperature of the warmer decreases and the temperature of the cooler increases.
- The energy exchange ceases when the objects reach thermal equilibrium.
- The concept of energy was broadened from just mechanical to include internal.
  - *Made conservation of energy a universal law of nature.*



# Heat Compared to Internal Energy

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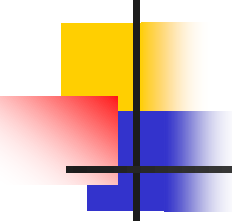
- Important to distinguish between them
  - *They are not interchangeable*
- They mean very different things when used in physics.



# Internal Energy

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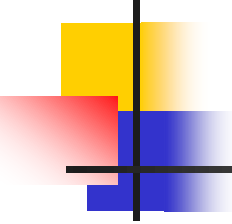
- *Internal energy,  $U$ ,* is the energy associated with the microscopic components of the system.
  - Includes kinetic and potential energy associated with the random translational, rotational and vibrational motion of the atoms or molecules.
  - Also includes any potential energy bonding the particles together.



# Heat

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- *Heat* is the transfer of energy between a system and its environment because of a temperature difference between them.
- The symbol  $Q$ , is used to represent the amount of energy transferred by heat between a system and its environment.

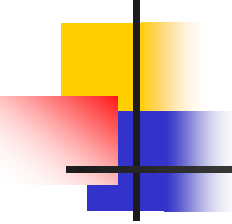


# Units of Heat

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## ■ Calorie

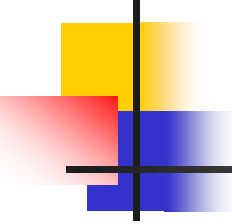
- A historical unit, before the connection between thermodynamics and mechanics was recognized.
- A *calorie* is the amount of energy necessary to raise the temperature of 1 g of water from 14.5° C to 15.5° C .
  - A *Calorie (food calorie)* is 1000 cal



# Units of Heat, cont.

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- 1 cal = 4.186 J
  - This is called the *Mechanical Equivalent of Heat*.

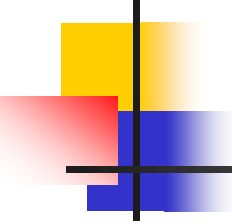


# Example 1.

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- A student eats a dinner rated at 2000 calories. He wishes to do an equivalent amount of work in the gymnasium by lifting a 50kg barbell. How many times must he raise the barbell to expend this much energy? Assume he raises the barbell 2m each time he lifts it and he regains no energy when he lowers the barbell.





# Solution

---

$$(1) \quad \Delta U_{\text{total}} = W_{\text{total}}$$

$$\Delta U = mgh$$

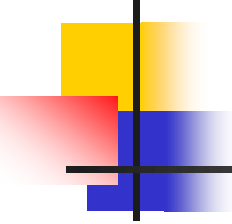
$$(2) \quad \Delta U_{\text{total}} = nmgh$$

$$nmgh = W_{\text{total}}$$

$$n = \frac{W_{\text{total}}}{mgh}$$

$$= \frac{(2\,000 \text{ Cal})}{(50.0 \text{ kg})(9.80 \text{ m/s}^2)(2.00 \text{ m})} \left( \frac{1.00 \times 10^3 \text{ cal}}{\text{Calorie}} \right) \left( \frac{4.186 \text{ J}}{1 \text{ cal}} \right)$$

$$= 8.54 \times 10^3 \text{ times}$$

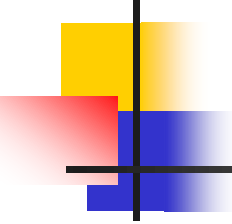


# Specific Heat

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- Every substance requires a unique amount of energy per unit mass to change the temperature of that substance by 1° C.
- The *specific heat, c*, of a substance is a measure of this amount.

$$c = \frac{Q}{m \Delta T}$$



# Units of Specific Heat

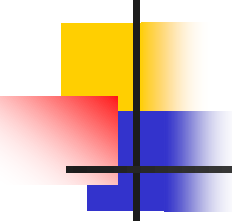
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- SI units
  - $\text{J} / \text{kg} \text{ } ^\circ\text{C}$

**TABLE 20.1***Specific Heats of Some Substances at 25°C and Atmospheric Pressure*

| Substance               | Specific Heat<br>(J/kg · °C) | Substance           | Specific Heat<br>(J/kg · °C) |
|-------------------------|------------------------------|---------------------|------------------------------|
| <i>Elemental solids</i> |                              | <i>Other solids</i> |                              |
| Aluminum                | 900                          | Brass               | 380                          |
| Beryllium               | 1 830                        | Glass               | 837                          |
| Cadmium                 | 230                          | Ice (−5°C)          | 2 090                        |
| Copper                  | 387                          | Marble              | 860                          |
| Germanium               | 322                          | Wood                | 1 700                        |
| Gold                    | 129                          | <i>Liquids</i>      |                              |
| Iron                    | 448                          | Alcohol (ethyl)     | 2 400                        |
| Lead                    | 128                          | Mercury             | 140                          |
| Silicon                 | 703                          | Water (15°C)        | 4 186                        |
| Silver                  | 234                          | <i>Gas</i>          |                              |
|                         |                              | Steam (100°C)       | 2 010                        |

*Note:* To convert values to units of cal/g · °C, divide by 4 186.

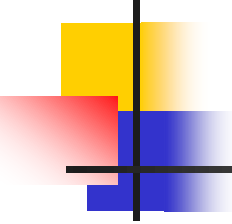


# Heat and Specific Heat

---

$$Q = m c \Delta T$$

- $\Delta T$  is always the final temperature minus the initial temperature.
  - *When the temperature increases,  $\Delta T$  and  $\Delta Q$  are considered to be positive and energy flows into the system.*
  - *When the temperature decreases,  $\Delta T$  and  $\Delta Q$  are considered to be negative and energy flows out of the system.*



## Example 2.

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Find the energy required to raise the temperature of 0.500 kg of water by 3.00°C.

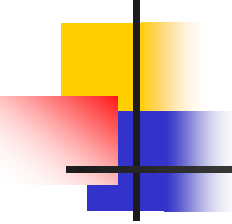
Solution:

From  $c \equiv \frac{Q}{m\Delta T}$  we get

$$Q = cm\Delta T$$

$$= (4\,186 \text{ J/kg}\cdot^\circ\text{C})(0.500 \text{ kg})(3.00^\circ\text{C})$$

$$= 6.28 \times 10^3 \text{ J}$$



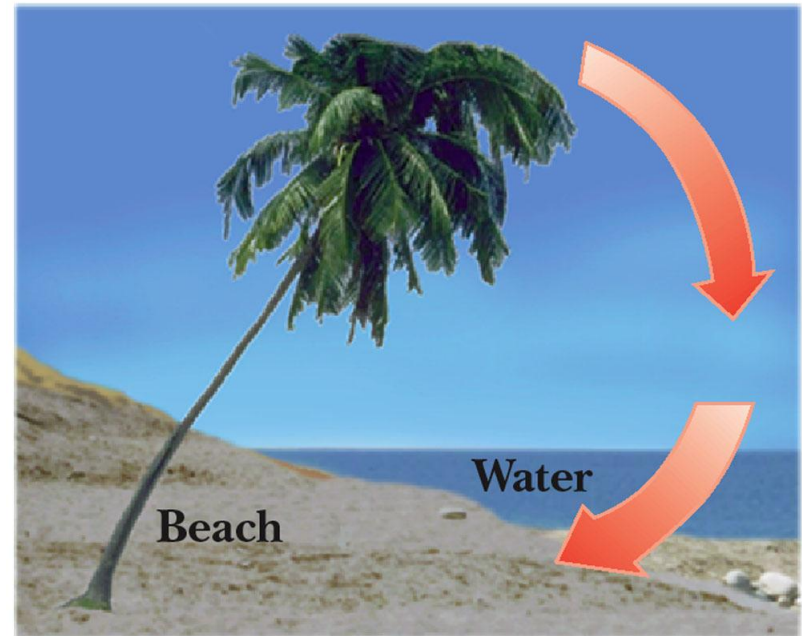
## Exercise 2.

---

1. The temperature of a silver bar rises by  $10.0^{\circ}\text{C}$  when it absorbs  $1.23\text{ kJ}$  of energy by heat. The mass of the bar is  $525\text{ g}$ . Determine the specific heat of silver.
2. A  $1.50\text{-kg}$  iron horseshoe initially at  $600^{\circ}\text{C}$  is dropped into a bucket containing  $20.0\text{ kg}$  of water at  $25.0^{\circ}\text{C}$ . What is the final temperature? (Ignore the heat capacity of the container, and assume that a negligible amount of water boils away).
3. A steel strut near a ship's furnace is  $2.00\text{ m}$  long, with a mass of  $1.57\text{ kg}$  and cross-sectional area of  $1.00 \times 10^{-4}\text{ m}^2$ . During operation of the furnace, the strut absorbs thermal energy in a net amount of  $2.50 \times 10^5\text{ J}$ .
  - i. Find the change in temperature of the strut
  - ii. Find the increase in length of the strut.

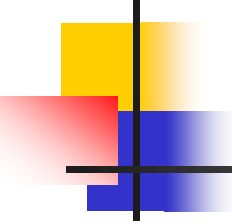
# A Consequence of Different Specific Heats

- Water has a high specific heat compared to land.
- On a hot day, the air above the land warms faster.
- The warmer air flows upward and cooler air moves toward the beach.



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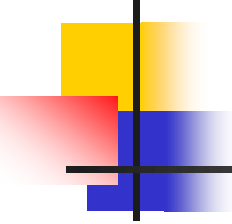




# Calorimeter

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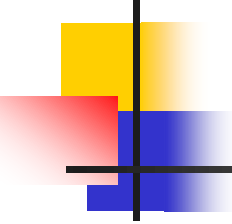
- One technique for determining the specific heat of a substance.
- A *calorimeter* is a vessel that is a good insulator which allows a thermal equilibrium to be achieved between substances without any energy loss to the environment.



# Calorimetry

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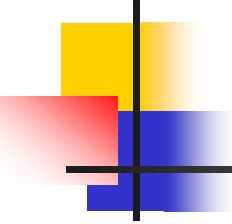
- Analysis performed using a calorimeter.
- Conservation of energy applies to the isolated system.
- ❖ *The energy that leaves the warmer substance equals the energy that enters the water.*
  - $Q_{\text{cold}} = -Q_{\text{hot}}$  i.e.  $m_w c_w (T_f - T_w) = -m_x c_x (T_f - T_x)$
  - Negative sign keeps consistency in the sign convention of  $\Delta T$



## Example 3.

---

A 125-g block of an unknown substance with a temperature of  $90.0^{\circ}\text{C}$  is placed in a Styrofoam cup containing 0.326 kg of water at  $20.0^{\circ}\text{C}$ . The system reaches an equilibrium temperature of  $22.4^{\circ}\text{C}$ . What is the specific heat,  $C_x$  of the unknown substance if the heat capacity of the cup is neglected?



# Solution

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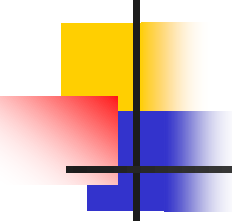
Let  $T$  be the final temperature, and let  $T_w$  and  $T_x$  be the initial temperatures of the water and block, respectively

Using equations:  $Q = cm\Delta T$  **and**  $Q_{cold} = -Q_{hot}$

*The water gains thermal energy  $Q_{cold}$ , while the block loses thermal energy  $Q_{hot}$*

$$Q_{cold} = -Q_{hot}$$

$$\therefore c_w m_w (T - T_w) = -c_x m_x (T - T_x)$$



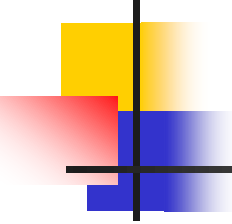
# Solution

---

Solving for  $c_x$  and substitute numerical values:

$$c_x = \frac{c_w m_w (T - T_w)}{m_x (T_x - T)}$$
$$= \frac{(0.326 \text{ kg})(4190 \text{ J/kg} \cdot ^\circ\text{C}(22.4^\circ\text{C} - 20.0))}{(0.125 \text{ kg})(90.0^\circ\text{C} - 22.4^\circ\text{C})}$$

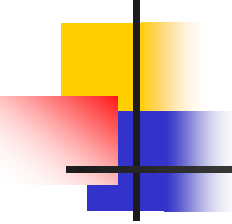
$$c_x = 388 \text{ J/kg} \cdot ^\circ\text{C}$$



## Exercise 3.

---

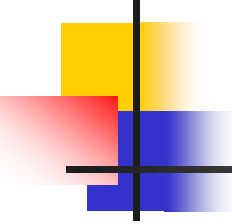
1. A 255-g block of gold at  $85.0^{\circ}\text{C}$  is immersed in 155 g of water at  $25.0^{\circ}\text{C}$ . Find the equilibrium temperature, assuming the system is isolated and the heat capacity of the cup can be neglected.
2. A 0.050 0-kg ingot of metal is heated to  $200.0^{\circ}\text{C}$  and then dropped into a beaker containing 0.400 kg of water initially at  $20.0^{\circ}\text{C}$ . If the final equilibrium temperature of the mixed system is  $22.4^{\circ}\text{C}$ , find the specific heat of the metal.



# Calorimetry with More Than Two Materials

---

- In some cases, it may be difficult to determine which materials gain heat and which materials lose heat.
- You can start with  $\Sigma Q = 0$ 
  - Each  $Q = m c \Delta T$
  - Use  $T_f - T_i$
  - *You don't have to determine before using the equation which materials will gain or lose heat.*

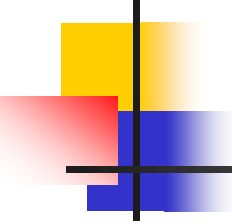


## Example 4.

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- A 0.050kg ingot of metal is heated to 200.0 °C and then dropped into a calorimeter containing 0.400kg of water initially at 20.0 °C. The final equilibrium temperature of the mixed system is 22.4 °C. Find the specific heat of the metal.





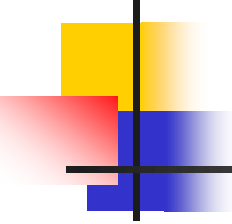
# Solution.

---

$$m_w c_w (T_f - T_w) = -m_x c_x (T_f - T_x)$$

$$c_x = \frac{m_w c_w (T_f - T_w)}{m_x (T_x - T_f)}$$

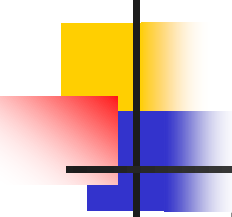
$$\begin{aligned} c_x &= \frac{(0.400 \text{ kg})(4186 \text{ J/kg} \cdot ^\circ\text{C})(22.4^\circ\text{C} - 20.0^\circ\text{C})}{(0.0500 \text{ kg})(200.0^\circ\text{C} - 22.4^\circ\text{C})} \\ &= 453 \text{ J/kg} \cdot ^\circ\text{C} \end{aligned}$$



# Phase Changes

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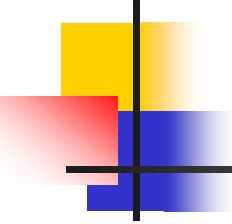
- A *phase change* occurs when the physical characteristics of the substance change from one form to another.
- Common phases changes are
  - Solid to liquid – melting
  - Liquid to gas – boiling
- Phases changes involve a change in the internal energy, but *no change in temperature*



# Latent Heat

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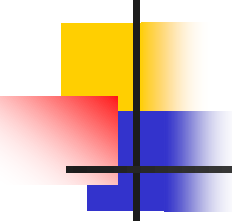
- During a phase change, the amount of heat is given as
  - $Q = \pm m L$
- $L$  is the latent heat of the substance
  - **Latent** means **hidden**
  - $L$  depends on the substance and the nature of the phase change.
- ❖ Choose a positive sign if you are adding energy to the system and a negative sign if energy is being removed from the system.



## Latent Heat, cont.

---

- SI units of latent heat are  $\text{J} / \text{kg}$ ,
- *Latent heat of fusion*,  $L_f$ , is used for melting or freezing.
- *Latent heat of vaporization*,  $L_v$ , is used for boiling or condensing,



# Sublimation

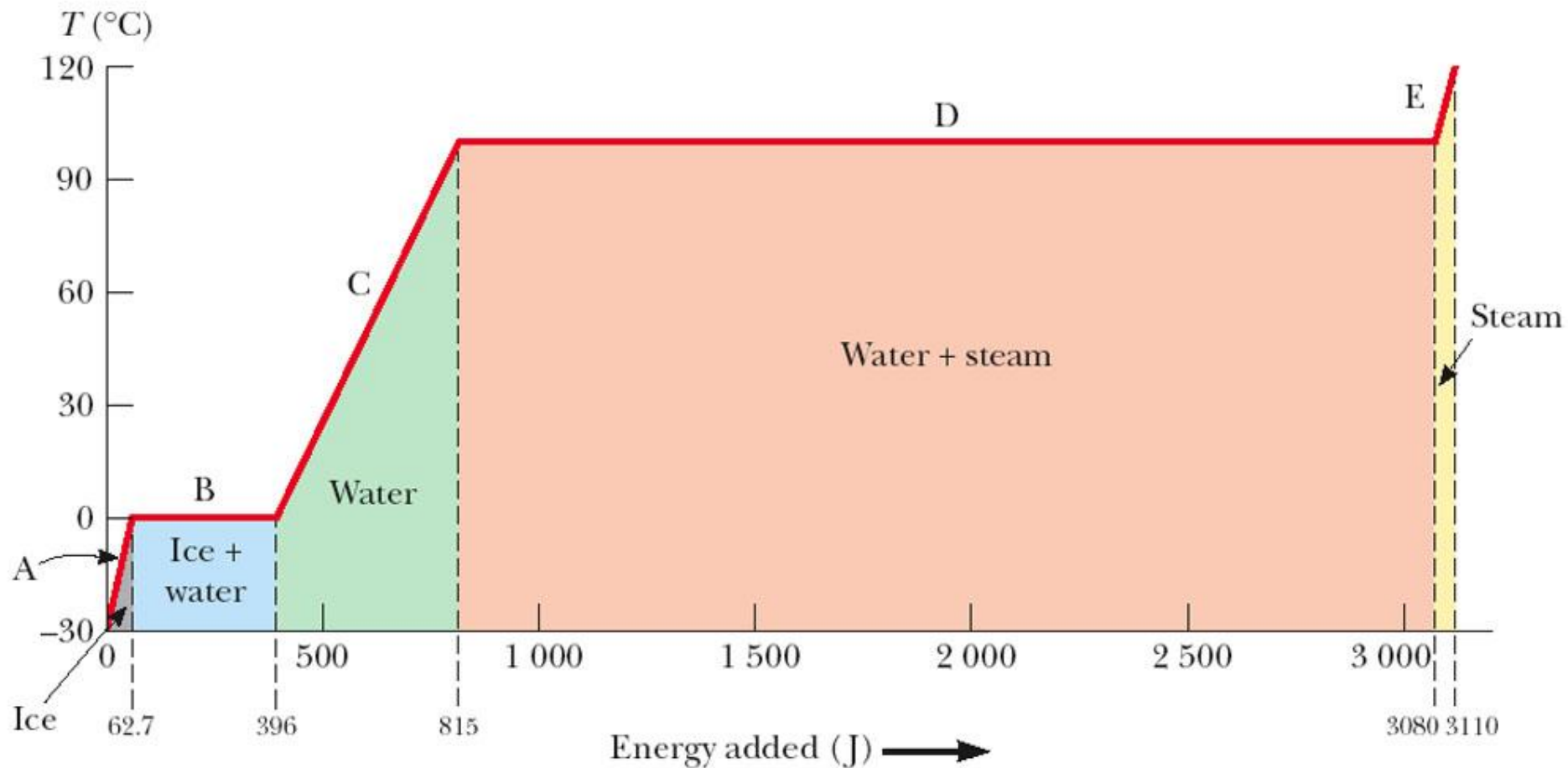
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- Some substances will go directly from solid to gaseous phase.
  - Without passing through the liquid phase.
- This process is called *sublimation*.
  - *There will be a latent heat of sublimation associated with this phase change.*
  - ***What substances undergo process of sublimation?***

**TABLE 20.2***Latent Heats of Fusion and Vaporization*

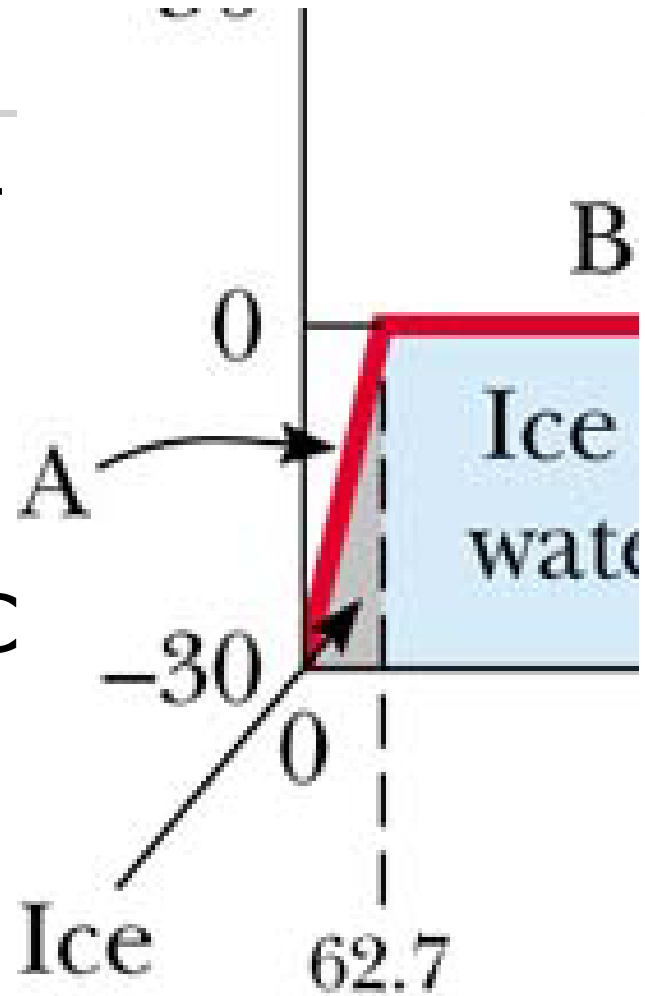
| Substance     | Melting Point (°C) | Latent Heat of Fusion (J/kg) | Boiling Point (°C) | Latent Heat of Vaporization (J/kg) |
|---------------|--------------------|------------------------------|--------------------|------------------------------------|
| Helium        | −269.65            | $5.23 \times 10^3$           | −268.93            | $2.09 \times 10^4$                 |
| Oxygen        | −218.79            | $1.38 \times 10^4$           | −182.97            | $2.13 \times 10^5$                 |
| Nitrogen      | −209.97            | $2.55 \times 10^4$           | −195.81            | $2.01 \times 10^5$                 |
| Ethyl alcohol | −114               | $1.04 \times 10^5$           | 78                 | $8.54 \times 10^5$                 |
| Water         | 0.00               | $3.33 \times 10^5$           | 100.00             | $2.26 \times 10^6$                 |
| Sulfur        | 119                | $3.81 \times 10^4$           | 444.60             | $3.26 \times 10^5$                 |
| Lead          | 327.3              | $2.45 \times 10^4$           | 1 750              | $8.70 \times 10^5$                 |
| Aluminum      | 660                | $3.97 \times 10^5$           | 2 450              | $1.14 \times 10^7$                 |
| Silver        | 960.80             | $8.82 \times 10^4$           | 2 193              | $2.33 \times 10^6$                 |
| Gold          | 1 063.00           | $6.44 \times 10^4$           | 2 660              | $1.58 \times 10^6$                 |
| Copper        | 1 083              | $1.34 \times 10^5$           | 1 187              | $5.06 \times 10^6$                 |

# Graph of Ice to Steam



# Warming Ice

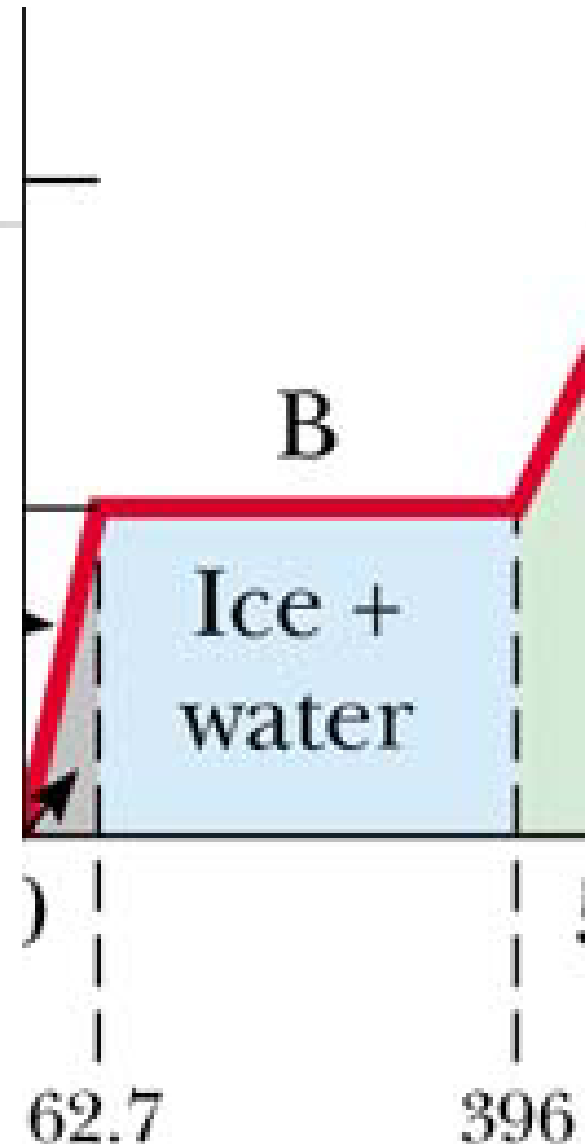
- Start with one gram of ice at  $-30.0^{\circ}\text{C}$
- During A, the temperature of the ice changes from  $-30.0^{\circ}\text{C}$  to  $0^{\circ}\text{C}$
- Use  $Q = m c \Delta T$
- Will add 62.7 J of energy





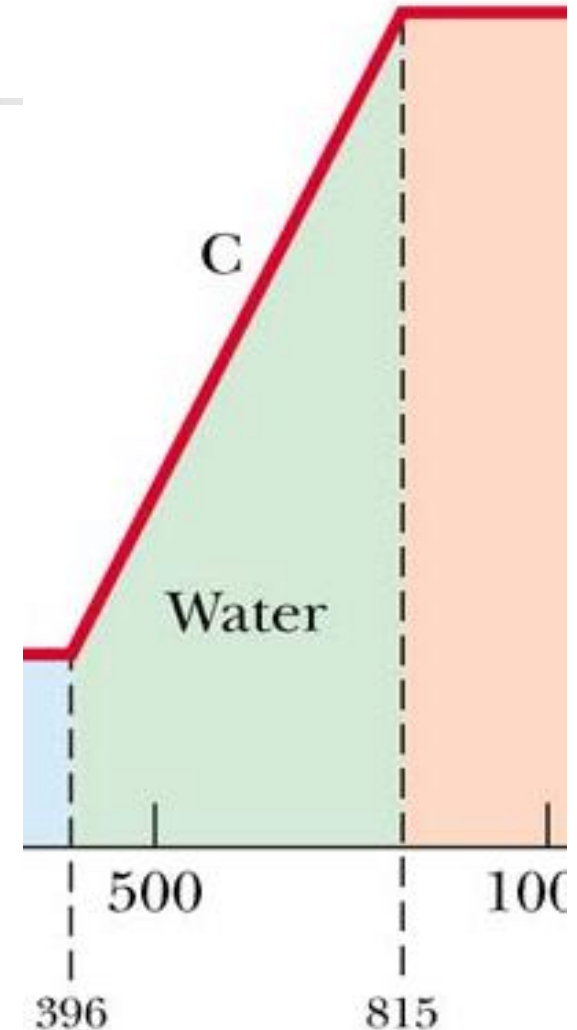
# Melting Ice

- Once at  $0^{\circ}\text{C}$ , the phase change (**melting**) starts.
- The temperature stays the same although energy is still being added.
- Use  $Q = m L_f$
- Needs 333 J of energy.



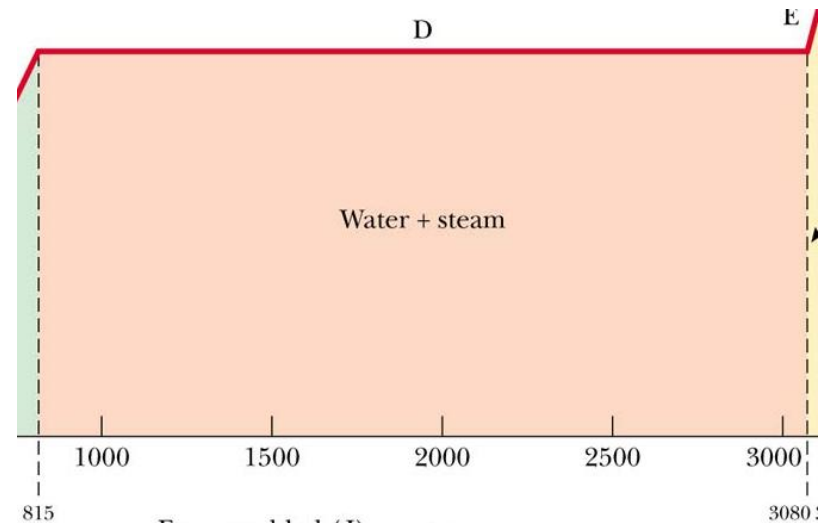
# Warming Water

- Between 0° C and 100° C, the material is liquid and no phase changes take place.
- Energy added increases the temperature.
- Use  $Q = m c \Delta T$
- 419 J of energy are added



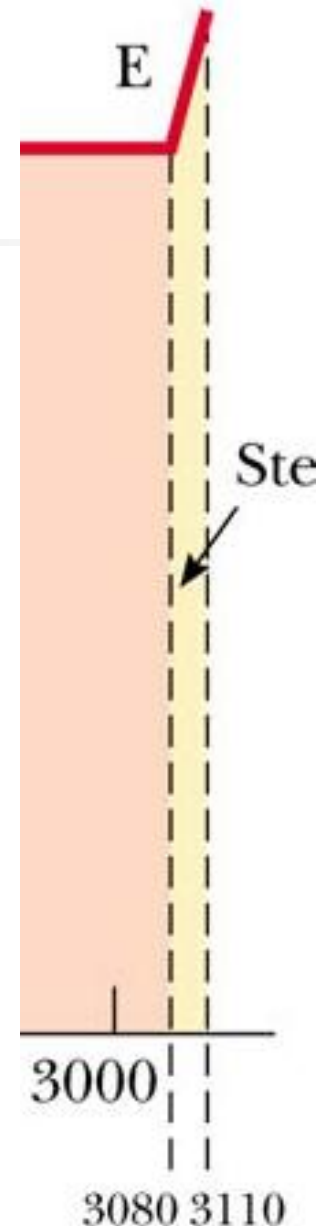
# Boiling Water

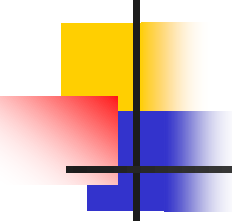
- At  $100^{\circ}\text{C}$ , a phase change occurs (boiling).
- Temperature does not change.
- Use  $Q = m L_v$
- 2 260 J of energy are needed



# Heating Steam

- After all the water is converted to steam, the steam will heat up.
- No phase change occurs.
- The *added energy goes to increasing the temperature.*
- Use  $Q = m c \Delta T$
- To raise the temperature of the steam to  $120^\circ$ , 40.2 J of energy are needed





# Calorimetry Problems

---

- If you have difficulty in solving calorimetry problems, be sure to consider the following points:
  - ✓ Units of measure must be consistent. For instance, if you are using specific heats measured in  $J/kg \cdot ^\circ C$ , be sure that masses are in kilograms and temperatures are in Celsius degrees.
  - ✓ Transfers of energy are given by the equation  $Q = cm\Delta T$  only for those processes in which no phase changes occur. Use the equations  $Q = \pm mL_f$  and  $Q = \pm mL_v$  only when phase changes are taking place; be sure to select the proper sign for these equations depending on the direction of energy transfer.
  - ✓ Often, errors in sign are made when the equation  $Q_{cold} = -Q_{hot}$  is used. Make sure that you use the negative sign in the equation, and remember that  $\Delta T$  is always the final temperature minus the initial temperature.

# Thank You For Your Attention !!!

